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Diagnostic Utility of Triple-Phase CT: Differentiating Benign Liver Lesions from Hepatocellular Carcinoma in Dogs

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ABSTRACT

This study aimed to identify triple-phase computed tomographic features that can predict the histotype of focal liver lesions in dogs. The analysis included dogs with histopathologically diagnosed nodular hyperplasia (NH, $n = 3$), hepatocellular adenoma (HCA, $n = 32$), or hepatocellular carcinoma (HCC, $n = 59$). Consistent with previous studies (Kutara et al. 2014; Burti et al. 2021), significant differences were observed in maximum transverse diameter ($p = .008$) and enhancement patterns ($p = .0031$) among the lesion types. HCCs were significantly larger and exhibited heterogeneous enhancement compared with benign lesions. In the portal venous phase, benign lesions were significantly hyperattenuating ($p < .001$) with a mean HU of 175.3 (±38.8), while HCCs were significantly hypoattenuating ($p < .001$) with a mean HU of 123.9 (±28.8), relative to the surrounding liver parenchyma (mean HU 151.6 ± 17.7). In the delayed phase, benign lesions became isoattenuating (mean HU 117.2 ± 10.7) to the liver parenchyma (mean HU 122.6 ± 8.4), whereas HCCs remained hypoattenuating (mean HU 100.3 ± 12.7). A maximal transverse diameter greater than 9.8 cm (AUC = 0.73), a heterogeneous enhancement pattern in all three phases (AUC = 0.7), and a portal venous phase HU below 136 (AUC = 0.87) were significantly associated with HCC, achieving an accuracy of 89% and a positive predictive value of 91%. The study suggests that lesion HU in the portal venous phase, alongside lesion size and enhancement pattern, are strong predictors of HCC in dogs, with specific cutoff values serving as reliable indicators.

1 | Introduction

Hepatic masses in dogs can result from various pathological conditions. Although metastatic liver diseases are observed in a higher percentage of cases, around one-third of dogs are presented with primary hepatic lesions [1–3].

Hepatocellular carcinoma (HCC) is the most common primary hepatic neoplasm in dogs [3–5], typically presenting as solitary or massive lesions [6, 7]. In addition to malignant lesions, benign hepatocellular adenomas (HCA) or hyperplastic nodular lesions (NH) of the liver parenchyma are also common [8–10]. In a study, 70% of dogs examined postmortem exhibited macroscopically visible NH [11]. Another study found that 27% of dogs with macroscopic liver and spleen masses had histologically benign lesions [12].

Differentiating between these primary liver masses is of paramount importance, as they require different therapeutic approaches, and HCCs have significantly different prognoses compared with HCAs and NHs. The advised course of action for dogs with massive HCCs is surgical resection. Nonsurgical treatment results in significantly shorter survival times (<270 days) compared with surgical treatment (>1460 days), with a 15.4 times higher tumor-related mortality rate in nonsurgically treated dogs [13].

Although histopathology and occasionally immunohistochemistry [14] are required for a definitive diagnosis, dynamic CT imaging has emerged as a valuable tool in differentiating benign from malignant neoplasms in both human [15, 16] and canine patients [17–19]. This imaging modality is not only essential for preoperative staging, surgical planning, and metastasis detection, but it also plays a crucial role in assessing the malignancy of the mass, providing a more comprehensive approach to diagnosis and treatment. The ability to capture computed tomographic images at the peak of enhancement of focal hepatic lesions is crucial for accurate detection. Multiphase CT protocols aim to delineate differences in the blood supply of lesions. For instance, hypervascular hepatic lesions, such as neoplasms with marked neovascularization, are primarily supplied by the hepatic artery and, therefore, exhibit pronounced contrast enhancement in early postcontrast CT images [20–22]. Conversely, hypovascular lesions, like metastases with necrosis, typically demonstrate diminished contrast uptake [18, 21]. However, it has been demonstrated that benign lesions, such as hepatic NH, can display contrast uptake patterns similar to those of malignant neoplasms when using dual-phase CT [21].

Triple-phase helical CT, its application, and its characteristics in dogs with various hepatic masses have been thoroughly explored with different outcomes. Some studies suggested that dual- or triple-phase CT imaging methods may not be able to differentiate between benign and malignant lesions [21, 23]. However, the majority of studies have identified specific CT features that can help predict the histological properties of focal liver lesions [17, 19, 24, 25]. These encompass an array of both quantitative CT parameters, such as the Hounsfield Units (HU) measured before the administration of contrast agents and the maximum transverse diameter, alongside qualitative attributes, including lesion surface features, morphological presentation, presence of encapsulation, and identification of any concomitant lymph node alterations. Based on these criteria, the overall accuracy in correctly assigning hepatic lesions ranged from 62% to 93.5% [17, 19, 25].

While prior studies established specific cutoff values for the size of the masses, indicating a higher likelihood of malignancy with increased size [17, 19], there is only limited access to cutoff values for the Hounsfield units (HU) of the mass in the respective contrast phases [25].

This study aims to address the existing knowledge gap by (1) identifying significant qualitative and quantitative CT features with strong interobserver agreement, (2) establishing cutoff values for these quantitative features to differentiate between HCA/NH and HCC based on a substantial retrospective case series, and (3) evaluating the diagnostic performance of these features in a prospective case series. We hypothesize that specific CT features and their cutoff values can effectively predict HCC.

2 | Materials and Methods

2.1 | Study Design and Study Population

This study was a single-center study conducted at the small animal hospital "Tierklinik Hofheim" in Germany (Katharina-Kemmler-Str.7, 65710 Hofheim am Taunus).

All the methods were carried out in accordance with the relevant guidelines and regulations. The data used in this study were part of routine clinical activity; therefore, no ethical committee approval was required.

The hospital database was queried for dogs referred between January 2019 and December 2023 with histopathologically confirmed hepatic lesions who received presurgical abdominal CT scans. Only cases of HCC, HCA, and NH were considered. The dogs included underwent bolus-tracked triple-phase CT scans comprising precontrast, arterial phase, portal venous phase, and delayed contrast series. Dogs were excluded from the study if they did not undergo triple-phase CT examination and did not have a definitive diagnosis of a lesion that was histopathologically confirmed.

The study comprised two components. First, a retrospective analysis of triple-phase CT features of liver masses aimed to identify notable distinctions between benign lesions (HCA, NH) and HCC. In the second phase, a prospective evaluation of triple-phase CT scans was undertaken, applying the criteria established in the retrospective phase to tentatively diagnose either HCC or HCA/NH. Subsequently, accuracy, sensitivity, and specificity were calculated to determine their capability of correctly identifying HCC and distinguishing it from benign lesions.

2.2 | Histopathological Examination

Histological samples were obtained either through ultrasound-guided Tru-Cut biopsy of the hepatic mass or through surgical removal of the mass. A pathologist carried out the histological examination at an external laboratory. In unclear cases, Ki67 immunohistochemistry was performed by the laboratory, with over 50% of Ki–67–positive cells being indicative of malignancy.

2.3 | Triple-Phase Helical CT Technique

All the animals were fasted for a 12 h period prior to examination. General anesthesia was always administered. All dogs were positioned in sternal recumbency during the scans.

The CT scans were performed using a 16 MDCT scanner (Aquilion RXL; Toshiba Medical Systems). The following scanning parameters were used: rotation time of 0.5–1 s, slice thickness of 1–2 mm, reconstruction interval of 0.5–1 mm, table speed of 1632 mm/rotation, helical pitch of 1.6, X-ray tube potential of 120 kV, and X-ray tube current of 250 mA. Iohexol (Accupaque 350; GE Healthcare Buchler GmbH & Co. KG) was administered as a contrast medium at a 2 mL/kg dose via the cephalic vein using a power injector. The injection speed was 3 mL/s (the range of injection time was 21–31 s). In patients with a body weight below 10 kg, a manual injection of the contrast agent was performed at the fastest possible rate.

Bolus tracking was performed on an aortic region of interest positioned at the diaphragm level, using a threshold of 180 HU to trigger the arterial scan. The scan delay for the arterial phase

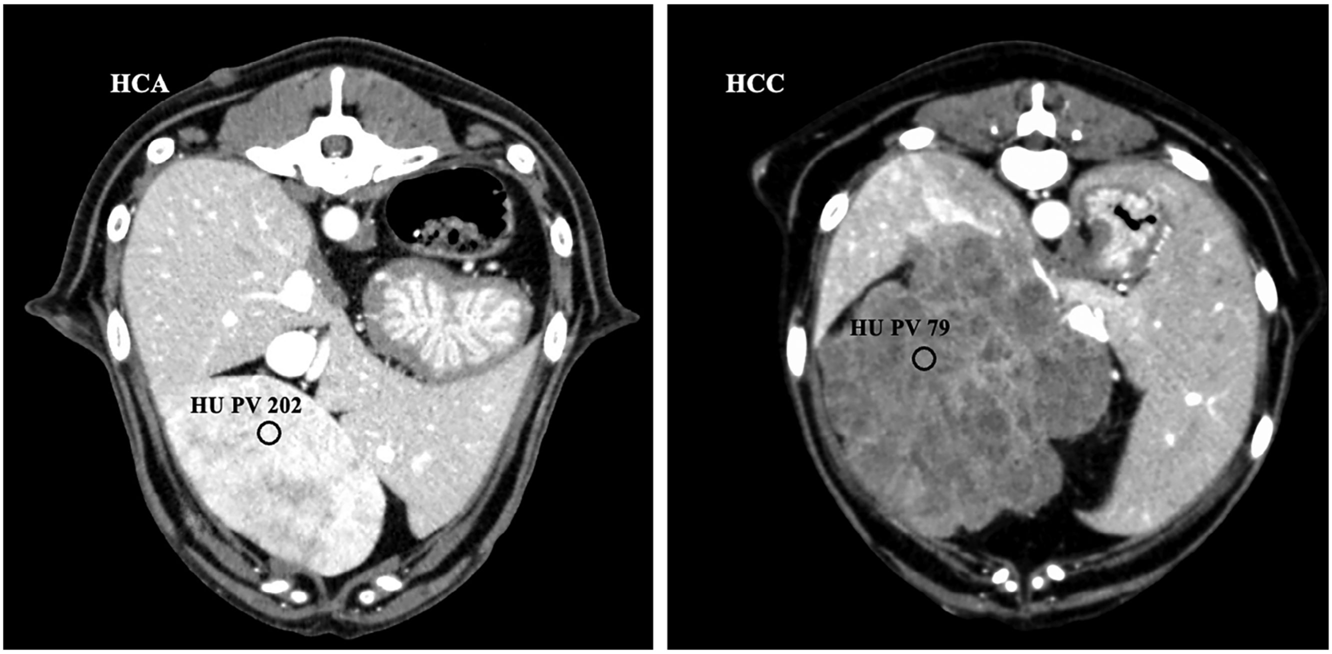

FIGURE 1 | Portal venous postcontrast computed tomographic images of the cranial abdomen in transverse views, comparing the Hounsfield units of hepatocellular adenoma (HCA, left side) and hepatocellular carcinoma (HCC, right side) with significantly higher values in HCA (202) than in HCC (97). Note the location of the ROI being placed in a parenchymatous and representative aspect of the mass. Also note the marked hyperattenuation of HCA compared with the surrounding liver parenchyma.

of opacification was 2 s after reaching the threshold. Portal venous and delayed phase images were then obtained 20 and 100 s after the initiation of the arterial scan, respectively.

2.4 | Image Analysis

In the retrospective evaluation, qualitative and quantitative CT features were recorded.

Quantitative parameters included the number of lesions, maximum transverse diameter, Hounsfield unit (HU) of the lesion, and liver parenchyma pre- and postcontrast application. For the measurement of the lesion HU, a solid/parenchymatous area was selected with a region-of-interest (ROI) of approximately 20 mm^2. The same location was chosen for the pre- and postcontrast phases. For postcontrast values, the most visually enhanced and representative portion of the lesion was used (Figure 1). The Hounsfield units of unaffected liver parenchyma were also measured in the same slice as the lesion or in an adjacent slice if necessary. Hyperattenuation was defined as a lesion contrast value at least 10 HU greater than the hepatic parenchyma contrast value. Isoattenuation was defined as a lesion contrast value within 10 HU of the hepatic parenchyma, while hypoattenuation was defined as a lesion contrast value at least 10 HU lower than that of the hepatic parenchyma.

Qualitative parameters included lesion location, margination (ill- and well-defined), surface (regular or irregular), contrast enhancement pattern (homogeneous vs. heterogeneous), mineralization, degree of cavitation/cyst-like areas (absent, mild, moderate, marked) within the lesion, distortion of the liver capsule, portal lymphadenomegaly, and abdominal effusion.

The scans were reviewed by two radiologists, one Diplomate ECVDI (J. W.) and a 1st-year resident (J. E.), using a picture archiving and communication system (PACS) workstation (OsiriX, version 13.0.3 Pixmeo SA). Both radiologists were blinded to the final diagnosis of the hepatic lesions. The qualitative and quantitative CT features were evaluated during the precontrast and postcontrast phases (arterial, portal venous, delayed). In the case of multiple lesions, only the CT features of the lesions that were sampled were described. If more than one lesion was sampled, the CT features of the largest lesion were evaluated. All the studies were displayed in a soft tissue window (WW: 400 HU—WL: 40 HU).

2.5 | Statistical Analysis

All the statistical analyses were performed using commercially available statistical analysis software (DATAtab Team, 2024). DATAtab: Online Statistics Calculator. DATAtab e.U. Graz, Austria. URL https://datatab.de).

The normality of quantitative data (lesion size, HU) was assessed by visual inspection and the Kolmogorov–Smirnov test. A *t*-test was performed to detect differences between HCC and benign lesions in normally distributed data. When data did not conform to a normal distribution, differences were tested using a Mann–Whitney *U*-test. HCAs and NH were evaluated collectively as benign lesions.

Differences in qualitative data (lesion margination, surface, mineralization, cavernosity, contrast enhancement pattern, distortion of liver capsule, portal lymphadenomegaly, abdominal effusion) were analyzed using Fisher's exact method. When cell frequencies were greater than 5, a Chi-squared test was used.

A Spearman's Rho (non-normally distributed data) or Pearson's correlation (normally distributed data) test was used to assess correlation for ordinal and continuous data.

Agreement between observers for quantitative and qualitative CT features was assessed using the intraclass correlation coefficient and Cohen's kappa, respectively.

For significant quantitative CT variables, the best cutoff value for predicting HCC was evaluated based on the highest Youden index and area under the curve (AUC) according to receiver operating characteristic curve analysis. A logistic regression analysis was performed to assess which combination of significant criteria was most useful in predicting HCC. An imaging criterion was considered useful to the diagnosis when it showed significant p-values and AUC-values above 0.7. A $p < .05$ was considered statistically significant. The diagnostic performance (sensitivity, specificity, positive and negative predictive values, accuracy) of each criterion was assessed individually and in combination.

The significant criteria were applied to a prospective dataset, and sensitivity, specificity, positive and negative predictive values, and accuracy were calculated. Accordingly, sensitivity in this study was defined as the proportion of dogs with histopathologically confirmed HCC whose tumors were correctly classified as HCC based on the CT criteria. Specificity was defined as the proportion of dogs with benign lesions on histopathology whose tumors were correctly classified as benign lesions based on the CT criteria. Diagnostic accuracy was defined as the proportion of dogs with either tumor that were correctly classified using the CT criteria.

3 | Results

3.1 | Retrospective Part

Thirty-six dogs met the inclusion criteria. The cases comprised 15 histopathologically confirmed HCAs and 21 confirmed HCCs, with immunohistochemistry required for differentiation in five cases (HCA: 2; HCC: 3).

Breeds included twelve small to medium-sized mixed breed dogs (8–23 kg; four of them below 10 kg), two Siberian Huskies, two Rhodesian Ridgebacks, two Dachshunds and one German Shepherd, Dalmatian, Chihuahua, Briard, Welsh Terrier, Pug, Eurasier, Podenco, American Staffordshire, Flat-Coated Retriever, Alaskan Malamute, Samoyed, Parson Russell Terrier, Jack Russell Terrier, Golden Retriever, Labrador Retriever, Australian Shepherd, and Border Collie. The mean age at the time of imaging was 13 years (range: 8–17 years). There were five intact males, eleven neutered males, three intact females, and seventeen neutered females.

The evaluation of seven qualitative and two quantitative variables was included in the analysis. Among these variables, one qualitative variable, the postcontrast enhancement pattern of the lesion, and the two quantitative variables, including maximal transverse diameter and HU of the lesion in the portal venous and delayed phase, were significantly different between benign and malignant liver lesions.

TABLE 1 | Quantitative characteristics of hepatocellular adenoma (HCA) and carcinoma (HCC) in the retrospective study cohort of animals.

	HCA	HCC	*p*-Value
Maximum transverse diameter (cm)	7.9 ± 4.2	11.4 ± 3.3	.008[a]
HU of liver parenchyma			
Before contrast	69.0 ± 5.8	67.4 ± 6.9	.465
Arterial phase	87.9 ± 8.4	89.7 ± 14.9	.987
Portal venous phase	160.7 ± 14.5	151.6 ± 17.7	.119
Delayed phase	122.6 ± 8.4	117.3 ± 8.5	.075
HU of the lesion			
Before contrast	62.5 ± 6.9	59.5 ± 6.2	.126
Arterial phase	109.4 ± 30.2	96.5 ± 35.9	.264
Portal venous phase	175.3 ± 38.8	123.9 ± 28.8	<.001[a]
Delayed phase	117.2 ± 10.7	100.3 ± 12.7	<.001[a]

Note: p-values < .05 were statistically significant.
Abbreviations: HU, Hounsfield units; HCA, hepatocellular adenoma; HCC, hepatocellular carcinoma.
[a]Statistical significance; values are presented as the mean ± standard deviation.

Quantitative features are summarized in Table 1. For quantitative features, a good to very good inter-observer agreement was detected.

HCC showed significantly larger lesions than HCA ($p = .008$; $t = -2.83$; 95% CI [−6.06, −0.99]; Cohen's $d = 0.96$). A cutoff value for maximum transverse diameter was set at 9.8 cm, with HCC being supposed to be larger than 9.8 cm (Youden index: 0.43, AUC = 0.73), correctly assigning 26 out of 36 lesions (~72%). There was no correlation between body weight and the maximum transverse diameter of the lesion.

For HU, HCA showed significantly higher values than HCC in the portal venous ($p < .001$; $t = 4.57$; 95% CI [28.49, 74.27]; Cohen's $d = 1.54$; Figure 1) and delayed contrast phase ($p < .001$; $t = 4.19$; 95% CI [8.55, 24.65]; Cohen's $d = 1.42$). Cutoff values for HU in the portal venous phase and delayed phase were determined with portal venous values below 136 (Youden index = 0.56; AUC = 0.87) and delayed values below 108 being indicative of HCC (Youden index = 0.66, AUC = 0.86), correctly assigning 29 out of 36 lesions (~81%) and 28 out of 36 lesions (~78%), respectively.

For qualitative features only for heterogeneity, cavernosity, portal lymphadenomegaly, and abdominal effusion, a substantial to very good interobserver agreement was detected. The other parameters had a fair to moderate interobserver agreement.

HCC showed significantly more often a heterogeneous contrast enhancement pattern in all three phases than HCA, with the delayed phase being the most significant phase for differentiating HCC from HCA ($p = .0031$; Youden index = 0.61; AUC = 0.7; Figure 2), correctly assigning 29 out of 36 lesions (~81%).

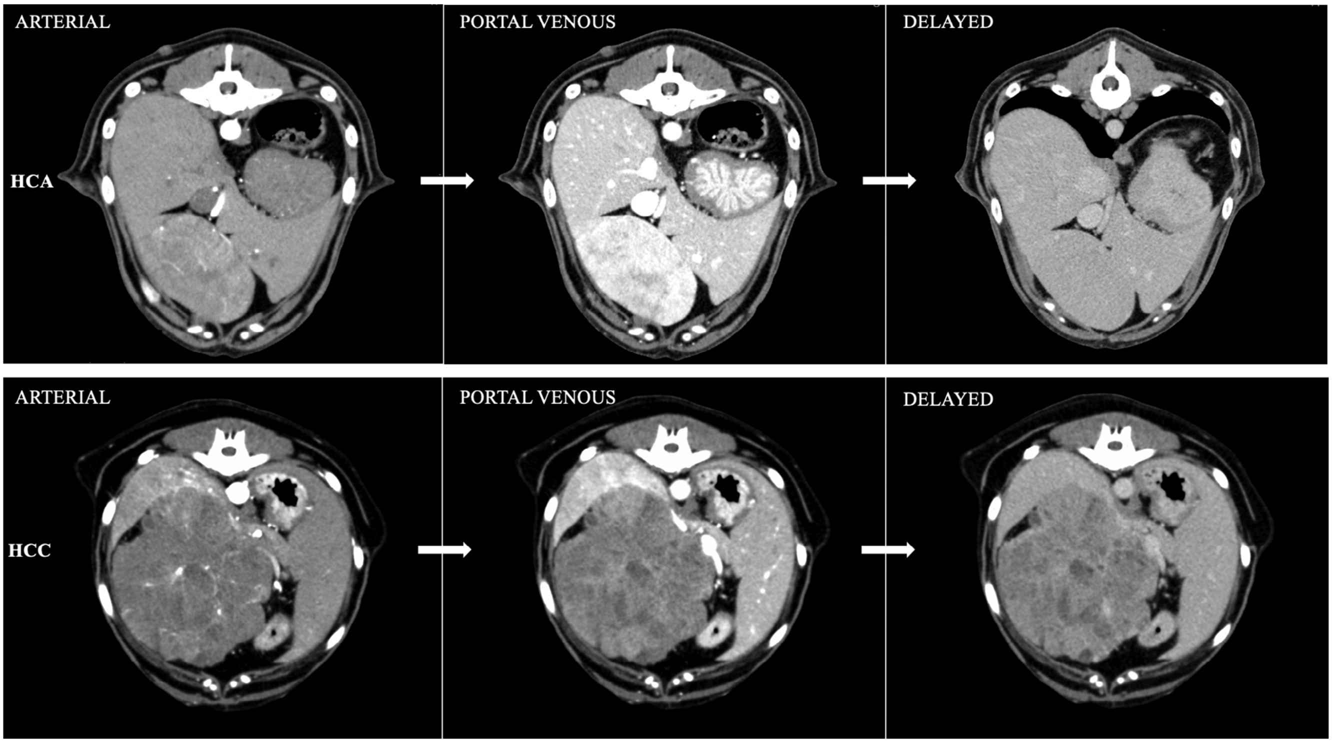

FIGURE 2 | Arterial, portal venous, and delayed postcontrast computed tomographic images of the cranial abdomen in transverse views, comparing the enhancement characteristics of hepatocellular adenoma (top row) and hepatocellular carcinoma (bottom row). Note the homogeneous enhancement pattern of the hepatocellular adenoma compared with the heterogeneous enhancement pattern of the hepatocellular carcinoma, being most pronounced in the portal venous and delayed phase.

Although higher grades of cavernosity were observed more often in HCC than in HCA, this finding was not statistically significant (Figure 3).

There was no significant difference in lesion margination, surface, or capsular distortion.

No mineralizations were detected in any of the lesions, irrespective of their biological behavior.

Portal lymphadenopathy was detected in 13 out of 21 HCCs and 6 out of 15 HCAs, without reaching a significant level. Only three animals (HCC: 1, HCA: 2) showed an abdominal effusion.

When testing for correlations, cavernosity showed a moderate, positive correlation with maximum transverse diameter ($r = 0.41$, $p = .014$) and a moderate, negative correlation with HU in the portal venous phase ($r = -0.36$, $p = .031$). Furthermore, HU in the portal venous phase showed a very high, positive correlation with HU in the delayed phase ($r = 0.8$, $p < .001$).

All significant CT variables from the univariate analysis were included in the logistic regression analysis. However, the model was run separately with either portal venous HU or delayed HU due to their strong positive correlation. The analysis revealed that the postcontrast enhancement pattern of the lesion in the delayed phase ($p = .012$; odds ratio = 19.36; 95% CI [2.01–186.6]), and portal venous HU ($p = .032$; odds ratio = 1.04; 95% CI [1–1.07]), as well as delayed HU ($p = .034$; odds ratio = 1.13; 95% CI [1.01–1.27]) of the lesion were found to be significant CT variables for the differentiation of benign from malignant (HCC) focal liver lesions. Maximum transverse diameter showed a p-value of .068 (odds ratio = 0.76; 95% CI [0.57–1.02]). However, it showed an AUC above 0.7 and was included to have a third variable being crucial for decision-making, as two out of three variables had to be fulfilled to decide on a lesion type. The diagnostic performance of each variable and the combination of variables for predicting HCC are summarized in Table 2. The results indicated that the combination of the variables had a higher predictive ability than each individual variable in predicting liver malignancy, with an area under the curve of 0.93–0.95. The combination of all three variables presented an overall accuracy of 88.9%.

3.2 | Prospective Part

Fifty-seven dogs were enrolled for the prospective part, including histopathologically confirmed 38 HCCs, 17 HCAs, and 2 NHs. Breeds included 16 small to large-sized mixed breed dogs (6–41 kg), three Australian Shepherds, two Labrador Retrievers, Pinschers, Beagles, Poodles, Yorkshire Terriers, Jack Russell Terriers, Border Collies, and one Rhodesian Ridgeback, Chow Chow, Chihuahua, Shetland Sheepdog, Tibetan Terrier, Irish Terrier, Airedale Terrier, Soft-Coated Wheaten Terrier, French Bulldog, Lagotto Romagnolo, Golden Retriever, and Bichon Frise. The mean age at the time of imaging was 13 years (range: 8–17 years). There were 8 intact males, 20 neutered males, 1 intact female, and 28 neutered females.

Using the predetermined criteria (maximum transverse diameter >9.8 cm, HU in the portal venous <136 and/or delayed phase

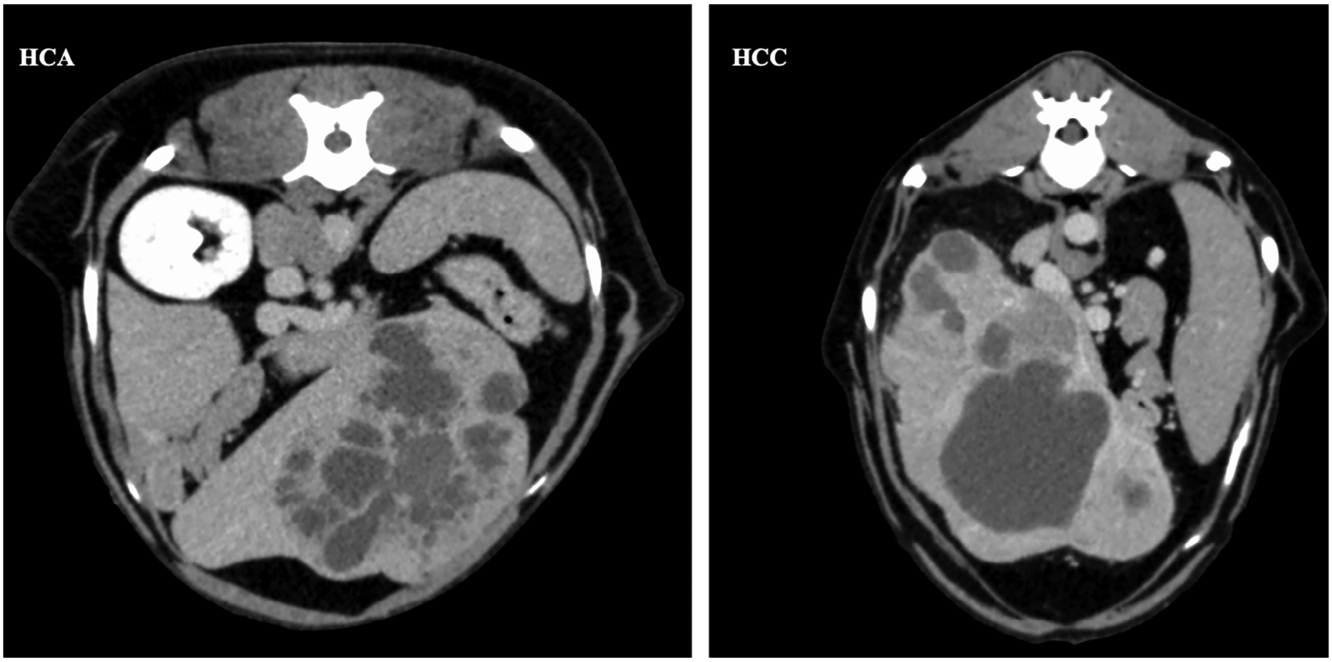

FIGURE 3 | Delayed postcontrast computed tomographic images of the cranial abdomen in transverse views, demonstrating a similar degree of cavernous or cyst-like areas of hepatocellular adenoma (HCA, left side) and hepatocellular carcinoma (HCC, right side).

TABLE 2 | Diagnostic performance of CT variables for predicting hepatocellular carcinoma.

	A. Hetero-geneity in all three phases	B. Maximum transverse diameter >9.8 cm	C. HU portal venous phase < 136	D. HU delayed phase < 108	Together: A, B, C	Together: A, B, D
Sensitivity	89.5	76.2	76.2	80	90.5	95.2
Specificity	71.4	66.7	80	85.7	86.7	80
PPV	80.9	76.2	84.2	88.9	90.5	87
NPV	83.3	66.7	70.6	75	86.7	92.3
Accuracy	81.8	72.2	77.8	82.4	88.9	88.9
Youden index	0.61	0.43	0.56	0.66	0.77	0.75
AUC	0.7	0.73	0.87	0.86	0.952	0.93

Abbreviations: HU, Hounsfield units; PPV, positive predictive value; NPV, negative predictive value; AUC, area under the curve.

<108, heterogeneous enhancement pattern in the delayed phase), 34 out of 38 HCCs and 15 out of 19 benign lesions were correctly identified. Overall, this led to an accuracy of 86%, sensitivity of 90%, and specificity of 79%, with a PPV of 90% and a NPV of 79%.

4 | Discussion

In this study, specific CT features of the liver lesion, such as a maximum transverse diameter exceeding 9.8 cm, portal venous HU below 136, delayed-phase HU below 108, and a heterogeneous enhancement pattern in the delayed phase, were significantly associated with HCC.

This aligns with prior research that identified features like maximum transverse diameter, contrast enhancement pattern, and HU thresholds for differentiating liver lesions [17, 19, 25].

The maximum transverse diameter threshold ranged from 9.5 cm [17] to 9.6 cm [25], closely aligning with the cutoff value identified in our study (9.8 cm). Conversely, a significantly lower cutoff (4.5 cm) was reported by [19]. This could be attributed to the predominance of NHs among their benign lesions (14 out of 18), as they are reported to be smaller than HCA [24]. Documented a median diameter of 8.55 cm for HCA, whereas NHs had a median diameter of 4.2 cm. HCAs in the present study had a median diameter of 7.9 ± 4.2 cm. Our study only examined three NHs with a mean diameter of 6.9 cm, potentially limiting its representativeness for this lesion type. However, since all NHs were below our threshold transverse diameter, there is no risk of lesion misclassification.

According to previous literature, the combination of CT features yielded higher accuracy rates than considering each individually. In the present study, the combination of previously mentioned features achieved an accuracy of around 86% with a positive predictive value for HCC of around 90%. This was significantly higher than the accuracy evaluated for each feature individually, ranging from 72% to 82%.

The accuracy of previous studies in correctly classifying hepatic lesions ranged from 62% to 93.5% [17, 19, 25]. Previous literature included a broad variety of malignant and benign liver lesions. The present study focused on the differentiation of HCC from HCA. In the prospective portion of the study, only a limited number of NH cases were evaluated. The authors assumed that the classification criteria for HCA could be applied to NH without risking misclassification. Previous literature has not indicated significant differences between NH and HCA in terms of contrast enhancement behavior or pattern, with the only notable distinction being the smaller size of NH. The sample size for HCA was small in previous literature, ranging from 2 to 14 cases [17, 21, 23, 24]. In contrast, our study included 31 cases. Previous studies often investigated lesion types consisting of small sample sizes and compared heterogeneous groups. While a study [17] differentiated between lesion types (HCC vs. NH vs. other benign vs. other malignant lesions) for their analysis, other studies [19, 21, 25] performed a binary classification (benign vs. malignant). A study [19], for example, analyzed other malignant lesions, such as hemangiosarcoma ($n = 1$) and cholangiocarcinoma ($n = 2$), together with HCC ($n = 43$). For benign lesions, a wide variety of types were analyzed together, such as inflammatory diseases, degenerative processes, and benign neoplasia.

The inclusion of diverse malignant lesions within a single category can impact the interpretation of enhancement patterns. A prior study in dogs suggested that arterial-phase rim enhancement and progressive enhancement may be characteristic of cholangiocellular carcinoma [26], aligning with findings reported in human studies [27, 28]. This enhancement pattern distinguishes cholangiocarcinoma from HCC, which typically exhibits hypoattenuation in portal venous and delayed phases. Hemangiosarcomas often display strong contrast enhancement with accumulation [21, 29], creating a pattern that further contrasts with the hypoattenuating appearance of HCC [17, 24]. Consequently, analyzing them together could significantly elevate the mean HU of malignant lesions and lead to insignificant differences when compared with the HU of benign lesions. Benign lesions, such as NH and HCA, often show iso- to hyperattenuation in the venous phase [18, 24, 17]. A study [24] analyzed the largest cohort of HCA among available studies and reported a higher frequency of hyperattenuation in both the arterial and portal phases compared with HCC, though the differences were not statistically significant. This is in accordance with the present study, as marked hyperattenuation was one of the major characteristic findings of HCA. Possibly due to having nearly double the number of HCA cases, this finding showed statistical significance in the present study. The precise etiology of the marked contrast enhancement in HCA remains undefined. In humans, HCC's enhancement pattern is influenced by both mass size and cellular differentiation [30–32]. Studies have found that poorly differentiated HCC tends to exhibit earlier washout, possibly due to fewer wide tumor sinusoids facilitating faster blood flow [33]. Additionally, [34] noted changes in drainage vessels during hepatocarcinogenesis, transitioning from hepatic veins to hepatic sinusoids and ultimately to portal veins during dedifferentiation. It has further been proposed that lesions devoid of hypervascular components may represent well-differentiated tumors in which the preexisting hepatic arteries have been obliterated by the neoplasm, yet neovascularization has not yet occurred [35].

This argument might explain why the literature shows heterogeneous results regarding the enhancement of HCC and possible reasons for the early washout of HCC. The present study could not detect a correlation between size and HU, and unfortunately, specific information on cellular differentiation was lacking in most of the histological reports.

Nevertheless, HU values in the portal venous phase were negatively correlated to the degree of cavernosity. Hepatocellular carcinomas have been described as cyst-like lesions with heterogeneous contrast enhancement, especially in the delayed phase [17–19, 24]. These features resemble those found in the present study; indeed, all (21/21, retrospective part) HCCs were cyst-like/cavernous lesions, and a heterogeneous distribution of the contrast medium in all three phases was evident in the majority of cases (16/21). Contrary to HCCs, HCAs primarily showed a homogeneous contrast enhancement in the portal venous and/or delayed phase (10/15). However, cyst-like areas were also detected in the majority of HCAs (12/15), although to a lesser extent than in HCCs. For HCCs, these cyst-like areas were theorized to represent a region of necrosis. The presence of necrosis within a tumor is strongly related to aggressive pathological characteristics and may be the result of tumor-related hypoxia as tumors outgrow their blood supply [36]. According to a study [24], cyst-like, necrotic lesions were less commonly observed in benign lesions like NH but were more frequently seen in HCAs. This aligns with the results of the present study. Therefore, the presence of necrosis in hepatocellular masses may be related to their size and not just their aggressive pathological features.

Based on these findings, the region for HU measurement should be selected carefully. Using the lowest attenuation to predict lesion type might be misleading, particularly for HCA. A study [25] conducted HU measurements within necrotic, cavernous areas and established a threshold of 37 HU in the delayed phase, achieving an accurate prediction of 91.3% of the masses as benign or malignant. The benign masses primarily included NHs, with no HCAs. Conversely, in the present study, the measurement was performed within parenchymatous, hyperattenuating parts, which were representative of the overall appearance and contrast enhancement of the lesion. This was the first time HU cutoff values of the lesion were determined for the portal venous (<136 HU) and delayed phase (<108) with an accuracy ranging from 81 % (retrospective part) to 86 % (prospective part) to predict HCCs. Recent literature has suggested that relying solely on HU for enhancement evaluation may lead to a misleading differential diagnosis of HCC in dogs due to variable enhancement patterns [18]. However, the same study reported mean portal venous and delayed HU values for HCC (120.9 ± 43.6, 104.6 ± 30.3, respectively), which would have been accurately classified using our cutoff values. Similarly, mean portal venous and delayed HU values for NH (155.6 ± 25.7, 126.2 ± 16.6, respectively) would have been correctly classified as a benign lesion with the assistance of our cutoff values.

It is important to note the significant variability in HU after contrast application, as indicated by a large standard deviation in the present study. Therefore, the authors agree on the necessity of using multiple features for accurate lesion-type prediction. A two-out-of-three rule was adopted, suggesting that the third variable can be crucial in cases where the other two variables are contradictory.

Quantitative data proved to be more reliable, exhibited better interobserver agreement than qualitative data, and allowed for the definition of cutoff values. Although lesion HU in the portal venous and delayed phase showed a strong positive correlation, cutoff values were defined for both. In clinical scenarios where no triple-phase was performed, and only a portal venous or delayed phase is available, it can be particularly helpful to have both cutoff values. Regarding the diagnostic performance, there was no marked difference between the usage of the lesion HU in the portal venous or delayed phase, and accuracy remained the same (~89%).

This study included only one qualitative feature, which was the contrast enhancement pattern across all three phases (homogeneous vs. heterogeneous). The qualitative feature was included due to its significant statistical results, improved diagnostic performance, and established relevance for malignancy in previous studies [17–19, 24].

Other qualitative features, such as nodular surface, capsular distortion, lymph nodes, and abdominal effusion, played a significant role in a minority of previous literature and showed no significant differences in the present study.

This study had some limitations. The constant rate of contrast medium application, which was used in all animals, has been shown to cause high variability in scanning times [37], influenced by the patient's body weight and cardiovascular status. Employing a fixed injection duration protocol could have yielded more consistent results, especially regarding HU measurements with reduced standard deviation [38]. Additionally, in four cases, the contrast bolus was administered manually, which may have contributed to increased variability in the results.

Another major limitation was that no other benign and malignant lesions were included in the present study, limiting its value in clinical application. However, the authors believed that detecting specific features of a lesion requires a large sample size and a homogeneous group rather than a wide variety of lesions with small sample sizes. Moreover, previous studies have established that HCC is the most frequently occurring malignant solitary liver lesion, while NH is the most common benign solitary lesion, followed by HCA [7, 39]. Additionally, it is widely acknowledged that distinguishing well-differentiated HCC from HCA and NH presents a significant challenge in cytology, often requiring immunohistochemistry [40]. Due to their cytological similarities, the authors assumed that significant differences between NHs and HCAs were minimal and, therefore, evaluated them together as benign lesions in the prospective part. None of the NHs were misclassified during the blinded evaluation. Nonetheless, future studies examining triple-phase CT features of other benign and malignant solitary as well as multifocal lesions are necessary to enhance the accuracy of radiologists' predictions.

Another limitation could be the placement of the ROI. The observer aimed to select a parenchymatous area of the lesion that was free of artifacts and exhibited strong contrast enhancement while avoiding cavernous, cyst-like regions, which likely represent necrotic tissue. The central area was avoided due to its predominantly necrotic nature, and the peripheral parts were avoided to prevent partial volume artifacts. This subjective selection process could result in variability among observers when using cutoff values for decision-making. However, the present study showed strong interobserver agreement in HU measurements. In clinical settings, this subjective placement may not always be feasible, as it requires clinical experience. In cases with significant cavernosity, measuring the HU of the lesion parenchyma and evaluating the enhancement pattern is challenging and may yield unreliable results. In such instances, the authors recommend basing decisions on lesion size and degree of cavernosity.

Although the findings of the present study primarily focus on the portal venous and delayed phases, the authors would like to emphasize the importance of triple-phase CT. First, it is suggested that the use of triple-phase helical CT may significantly improve the detection rate of hepatic metastatic lesions [18]. Second, the exact timing for each phase is very important in order to detect the peak of enhancement. This is crucial for the HU measurements and when working with cutoff values. A previous study [21] performed dual-phase CT and was not able to detect significant CT differences between benign and malignant liver lesions.

In summary, three triple-phase CT features helped significantly to differentiate hepatocellular carcinoma from hepatocellular adenoma and nodular hyperplasia, including a heterogeneous enhancement pattern in the delayed phase, HU below 136 in the portal venous, below 108 in the delayed phase, and a maximum transverse diameter above 9.8 cm. These hepatocellular carcinoma features showed a positive predictive value of approximately 90%, with an overall accuracy of 86% in a prospective evaluation.

This study was able to present specific triple-phase CT features of hepatocellular carcinoma in dogs, with specific cutoff values serving as reliable indicators, ultimately improving the prediction of histopathologic classification of solitary liver lesions. Large-scale studies are needed in order to determine the diagnostic sensitivity and specificity of these triple-phase CT characteristics when including other benign and malignant lesion types.

Author Contributions

Category 1

(a) Conception and design: Wennemuth, Einwaller.

(b) Acquisition of data: Wennemuth, Einwaller.

(c) Analysis and interpretation of data: Wennemuth, Einwaller.

Category 2

(a) Drafting the article: Einwaller.

(b) Reviewing article for intellectual content: Wennemuth.

Category 3

(a) Final approval of the completed article: Wennemuth, Einwaller.

Category 4

(a) Agreement to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved: Wennemuth, Einwaller.

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Conflicts of Interest

The authors declare no conflicts of interest.

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No reporting checklist was used.

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